

RED SEAL EXAM PREP

421A Heavy Duty Equipment Technician

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
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Mock Exam — answer all questions, then check the key

Q1. A starter motor cranks slowly even though the battery is fully charged. The MOST likely cause is:

- A) High resistance in the starter control circuit
- B) Voltage drop in the battery cables
- C) Starter relay stuck open
- D) Low alternator output voltage

Q2. In the correct order, the four strokes of a diesel engine cycle are:

- A) Intake, Compression, Power, Exhaust
- B) Power, Intake, Compression, Exhaust
- C) Compression, Intake, Power, Exhaust
- D) Intake, Power, Compression, Exhaust

Q3. A work-light circuit is dead. The technician removes the ISO mini relay and measures 85 ohms between terminals 85 and 86. What does this reading confirm, and what remains untested?

- A) Coil is shorted — resistance should read near zero when healthy
- B) Coil is good — the switched contacts 30 to 87 still need testing
- C) Contacts are good — the coil still needs a supply voltage check
- D) Relay is fully proven — the fault must be in the socket wiring

Q4. In a machine with a pressure-compensated variable displacement pump, the pump maintains 3,500 PSI system pressure even when no hydraulic functions are being used. What is happening, and is this normal?

- A) Normal — a pressure-compensated pump de-strokes to near-zero flow while holding its compensator setting pressure
- B) The pump should drop to a low standby pressure of 200–300 PSI — full pressure at standby always means a stuck compensator
- C) The pump relief valve is opening — this is the correct standby state for a pressure-compensated system
- D) Normal — the pump charges an accumulator during standby and releases when functions are activated

Q5. An articulated dump truck's transfer case is shifted to low range for steep downhill work, but the machine accelerates uncontrollably on the descent despite the operator applying the retarder. What is the most likely cause?

- A) The transfer case did not fully engage low range — the machine is actually operating in high range
- B) The retarder system is designed for uphill use only — hydraulic retarders are ineffective on descents exceeding 12%
- C) Transfer case low range provides more torque to the wheels — more torque causes the machine to accelerate downhill
- D) The inter-axle differential lock was not engaged before the descent — unequal axle speeds are causing the retarder to lose effectiveness

Q6. A shop is converting its excavator buckets to a hammerless GET retention system such as Cat Advansys. Compared with driving hammered retainer pins, what is the main safety advantage for technicians?

- A) Every tooth tip lasts twice as long between changes
- B) It removes hammer blows and flying-steel hazards
- C) Adapters no longer require periodic inspection
- D) Teeth can be changed with the bucket fully raised

Q7. During an in-frame overhaul of a wet-sleeve diesel engine, the coolant side of the cylinder liners shows deep, clustered pitting. What caused this damage?

- A) Acid etching from combustion gas leaking past the head gasket
- B) Electrolysis from stray electrical current in the cooling system
- C) Cavitation erosion from liner vibration and depleted coolant SCA
- D) Scale deposits from hard water used to top up the coolant level

Q8. A hydraulic oil sample is reported as ISO 4406 cleanliness code 22/20/17. What does this three-number code represent, and is this acceptable for a servo-hydraulic system with tight-tolerance components?

- A) The code indicates the oil has passed cleanliness testing — higher numbers indicate cleaner oil
- B) ISO 4406 only applies to transmission oils — hydraulic systems use a different cleanliness standard
- C) The three numbers represent viscosity, oxidation level, and water content — 22/20/17 is within normal limits for any hydraulic system
- D) Particle count ranges at 4µm, 6µm, and 14µm — heavily contaminated oil, not acceptable for servo systems

Q9. A heavy equipment air brake system has a low air pressure warning light that activates at 80 psi but the system should have 120 psi. A compressor cycle test shows the compressor builds from 100 to 120 psi in 30 seconds and cuts out at 120 psi. What component is MOST likely leaking?

- A) Air dryer — moisture bypass is diluting the air supply
- B) Air compressor — cannot maintain pressure after cut-out
- C) Governor — cutting out prematurely at 120 psi
- D) A system air leak downstream of the compressor

Q10. A machine is equipped with HEUI (Hydraulic Electronic Unit Injectors). Unlike common rail injectors, HEUI injectors use engine oil pressure to amplify fuel injection pressure. If the injection actuation pressure control (IAPC) circuit fails and oil pressure drops to zero, what happens to fuel injection?

- A) Fuel injection continues at a lower pressure — the injector uses residual fuel pump pressure directly
- B) Injection ceases entirely — HEUI injectors need high-pressure oil to actuate the intensifier piston
- C) The fuel system switches to a backup mechanical injection mode automatically
- D) Injection pressure increases — without oil pressure, the fuel pressure compensates by rising above normal

Q11. A wheel loader with hydraulic-boosted service brakes has a spongy pedal that requires excessive travel to achieve normal braking. Fluid level is full and no external leaks are present. What is the MOST likely cause?

- A) Master cylinder primary cup has deteriorated — fluid bypasses the seal
- B) Air entrained in the hydraulic brake system circuit
- C) Hydraulic boost pressure is too low — booster assist is inadequate
- D) Brake disc/pads are worn beyond service limits

Q12. A machine with a hydrostatic drive system loses tractive effort in both forward and reverse but the engine power appears normal. The MOST likely cause is:

- A) The variable displacement pump swashplate is not responding to operator input — stuck at minimum displacement
- B) The travel motor has seized — blocking flow in the circuit
- C) Transmission oil temperature is too low, causing high viscosity
- D) Low charge pump pressure causing the hydrostatic loop to cavitate

Q13. A Class D fire involving burning magnesium swarf is discovered in the shop. What is the correct extinguishing method?

- A) CO₂ extinguisher — smothers the fire by displacing oxygen
- B) A Class D dry powder (graphite or copper-based) extinguisher
- C) Class B dry chemical — the powder coats and suffocates the magnesium fire
- D) Large volume of water to rapidly cool the magnesium below ignition temperature

Q14. A proportional directional control valve is controlled by a PWM signal from the machine's ECM. The valve has a dither signal applied to it. What is the PURPOSE of the dither signal and what happens if it is absent?

- A) Dither increases the maximum flow rate through the valve — without dither, flow is restricted to 50% of rated capacity
- B) A small high-frequency signal that keeps the spool in constant micro-motion; without it the spool sticks, causing jerky response
- C) Dither compensates for temperature changes in the hydraulic oil — it adjusts valve gain automatically as oil viscosity changes
- D) Dither is a safety feature that returns the valve to center position if the ECM signal is lost

Q15. A wheel loader equipped with oil-cooled wet disc brakes develops brake drag after a brake system repair. The repair included replacing brake cooling oil seals but not the disc packs. What is the MOST likely cause of the drag?

- A) The brake cooling oil pressure is too high — exceeding 20 PSI will apply the brake discs
- B) The brake cooling oil is too viscous, creating drag between the spinning and stationary discs
- C) Residual apply pressure is keeping the pistons clamped, or the piston return spring is damaged or missing
- D) Wet disc brakes inherently drag slightly — this is normal and acceptable

Q16. An operator reports a sweet odour inside the cab, a greasy film that keeps forming on the inside of the windshield, and a damp floor mat below the HVAC unit. Which component is the most likely cause?

- A) A leaking heater core
- B) A failed blend door actuator
- C) A restricted cab fresh-air filter
- D) A leaking evaporator core

Q17. What is the purpose of the EGR (Exhaust Gas Recirculation) system on a modern diesel engine?

- A) To reduce NOx emissions by lowering peak combustion temperature
- B) To increase engine power by recirculating exhaust gases for a second combustion cycle
- C) To preheat the intake air during cold starts
- D) To improve fuel economy by recovering exhaust energy

Q18. A technician is matching a new hydraulic attachment to a carrier's auxiliary circuit before installation. Which pair of specifications must be compared against the attachment's requirements first?

- A) Pilot supply pressure and joystick detent holding force
- B) Pump case drain size and return filter micron rating
- C) Auxiliary circuit flow rate and relief pressure setting
- D) Hydraulic cooler capacity and reservoir oil volume

Q19. What does a hydraulic accumulator store?

- A) Pressurized hydraulic fluid held under a gas pre-charge
- B) A reserve of filtered hydraulic fluid only
- C) Pilot pressure for proportional valve control
- D) Hydraulic fluid contamination for scheduled disposal

Q20. A cracked bucket linkage must be arc-welded while still mounted on a machine that has several electronic control modules. Which precaution protects the machine's electronics?

- A) Set the welder to AC output so no DC can enter the machine wiring
- B) Run the engine at idle so the alternator absorbs stray weld current
- C) Disconnect the batteries and clamp the work lead close to the weld
- D) Ground the work lead to the counterweight at the far end of the frame

Q21. What is the purpose of a "final drive" reduction on heavy equipment?

- A) To reduce output speed and multiply torque at the drive wheels
- B) To increase travel speed at the expense of pulling force
- C) To filter transmission fluid before it reaches the drive axle
- D) To disconnect the transmission from the drive wheels during engine braking

Q22. A technician is diagnosing an intermittent no-start condition. The engine cranks but does not start, and the problem occurs randomly. All fuel and compression tests are normal. A scan tool shows the engine speed sensor (MPU) has an intermittent signal. What is the MOST likely root cause?

- A) Engine speed sensor has failed internally — replace sensor
- B) Chafed wiring or connector corrosion in the CKP sensor circuit
- C) ECM is failing to process the CKP signal correctly
- D) Sensor air gap is incorrect — engine vibration causes signal dropout

Q23. A hydraulic cylinder with a bore diameter of 4 inches is subjected to 2,000 PSI. What is the approximate force output?

- A) 6,280 lbs
- B) 8,000 lbs
- C) 25,133 lbs
- D) 50,000 lbs

Q24. A hydraulic system has normal pump pressure but SLOW cylinder movement. The MOST likely cause is:

- A) Restricted flow control valve or filter
- B) A worn hydraulic pump
- C) Air in the hydraulic reservoir
- D) Relief valve pressure set too high

Q25. A CAN bus network on a machine has a measured resistance of 60 ohms between CAN High and CAN Low with the network powered off. What does this indicate?

- A) Normal — the bus termination resistance is correct
- B) One of the two terminating resistors has failed open
- C) A short circuit exists between CAN High and CAN Low
- D) The CAN bus is missing both terminating resistors

Q26. A hydraulic accumulator is used in heavy equipment systems primarily to:

- A) Filter hydraulic fluid
- B) Store pressurized fluid for peak demand
- C) Cool the hydraulic fluid during operation
- D) Convert hydraulic pressure to electrical energy

Q27. A turbocharged diesel engine exhibits excessive black smoke at light loads and hesitation when the throttle is quickly applied. Under hard acceleration the smoke clears and power is adequate. What is the most likely cause?

- A) Injectors are worn and delivering excess fuel at all loads — full power at WOT confirms the pump is adequate
- B) Turbocharger response lag — insufficient boost at low RPM causes a rich condition until boost builds
- C) EGR valve stuck closed — recirculation gases are being replaced by excess raw fuel
- D) Fuel injection timing is advanced — early injection causes smoke at light loads but clears under load

Q28. When performing a torque wrench calibration check, the technician finds the torque wrench reads 10% low compared to the certified torque tester. What should be done?

- A) Add 10% to all torque specifications when using this wrench — the offset is acceptable
- B) Remove the wrench from service for recalibration or replacement
- C) Use the wrench for non-critical fasteners only
- D) Adjust the torque wrench by loosening the handle end by 10%

Q29. What is the primary function of a differential in a heavy equipment drive axle?

- A) To increase ground clearance
- B) To allow drive wheels to turn at different speeds
- C) To increase the gear ratio for more pulling power
- D) To disengage drive power during braking

Q30. A technician needs to replace a damaged section of 10-gauge wire in a circuit that carries 25 amps continuously. The only wire available is 14-gauge. Why is 14-gauge wire INAPPROPRIATE for this repair?

- A) 14-gauge wire has lower resistance than 10-gauge, which would increase current draw and damage the component
- B) 14-gauge wire has a lower ampacity than 10-gauge and would overheat on a 25-amp circuit
- C) 14-gauge wire is only rated for AC circuits — DC circuits require 10-gauge minimum
- D) The gauge number difference (14 vs 10) exceeds the maximum allowable splice ratio of 1.5 per circuit repair standard

Q31. A diesel engine has good power under light loads but experiences power loss, black smoke, and excessive exhaust back pressure under heavy load. The MOST likely cause is:

- A) Worn injector tips causing poor fuel atomization
- B) Low fuel rail pressure under load
- C) A clogged DPF restricting exhaust flow
- D) An EGR valve stuck open at high load

Q32. A hydraulic system experiences aeration (air in oil). Which symptom is MOST characteristic of aeration vs. cavitation?

- A) Aeration causes a high-pitched whining; cavitation causes foamy oil
- B) Aeration causes foamy oil; cavitation causes high-pitched noise
- C) Both conditions cause identical symptoms
- D) Aeration only affects cylinder speed; cavitation only affects pump pressure

Q33. During downhill dynamic retarding on an electric-drive haul truck without energy storage, an apprentice hears the grid blower running and asks where the braking energy goes. That energy is:

- A) Returned through the alternator to charge the batteries
- B) Fed back through the inverter to motor the engine
- C) Absorbed by the oil circuit of a driveline retarder
- D) Converted to heat in the retarding resistor grids

Q34. In a planetary gear set where the RING GEAR is the input, the PLANET CARRIER is held (stationary), and the SUN GEAR is the output — what type of motion results?

- A) Speed reduction with same direction rotation
- B) Speed increase (overdrive) with rotation reversal
- C) The system locks up — no output is possible with carrier held
- D) 1:1 ratio — no gear change

Q35. A first-year apprentice is asked to recover R-134a from a dozer's cab AC system at a shop in Canada. What must the apprentice hold before performing this task?

- A) A federal transport permit for moving pressurized service cylinders
- B) A written manufacturer directive approving the recovery procedure
- C) A recognized refrigerant handling environmental awareness certificate
- D) A provincial hoisting licence for the class of equipment involved

Q36. What does a turbocharger BOOST PRESSURE that is consistently LOWER than specification indicate?

- A) Intercooler is blocked
- B) Turbocharger shaft is over-spinning
- C) Fuel injectors are delivering too much fuel
- D) Intake restriction, boost leak, or worn turbo

Q37. Which symptom would MOST likely indicate a slipping transmission clutch pack?

- A) Torque converter overheating only
- B) Transmission makes grinding noise in all gears
- C) Engine RPM rises but machine speed does not
- D) Machine moves fine in forward but not in reverse

Q38. A starter motor draws NORMAL current but cranks very slowly. The MOST likely cause is:

- A) Starter solenoid contacts worn
- B) Faulty starter motor brushes
- C) High resistance in the battery cables
- D) Battery has too many cold cranking amps (CCA)

Q39. A diesel engine produces black smoke under full load but white smoke during warm-up that clears when hot. The fuel system and turbo are confirmed serviceable. What is the MOST likely cause of the black smoke under full load?

- A) Air filter restriction — insufficient air for complete combustion at full load
- B) Engine is operating normally — black smoke at full load is acceptable for older diesels
- C) Injector timing advanced — fuel injected too early causing incomplete combustion
- D) Coolant leak into combustion chamber — white smoke becomes black as load increases

Q40. A quick-release valve is installed close to the brake chambers. Its function is to:

- A) Balance pressure between front and rear axle brakes
- B) Provide a secondary pressure source for emergency braking
- C) Limit maximum brake application pressure
- D) Exhaust brake air rapidly at the chamber to reduce brake release time

Q41. What is the key operational difference between an open-center and a closed-center hydraulic system, and how does it affect fuel consumption?

- A) Open-center is more efficient because continuous flow prevents pump cavitation, reducing overall energy use
- B) Closed-center systems require a larger pump — the increased pump size consumes more fuel than an open-center design
- C) Open-center systems use hydraulic oil with an open (unsealed) reservoir; closed-center systems use a sealed pressurized reservoir — no difference in fuel consumption
- D) Open-center systems circulate oil to tank in neutral; closed-center systems stop pump flow at standby, saving energy

Q42. Under WHMIS 2015, what does the GHS pictogram showing a flame over a circle represent?

- A) Explosive material that can detonate under heat, shock, or friction
- B) Acute toxicity — fatal or toxic after a single exposure
- C) Flammable liquid — liquids with a flash point below 60°C
- D) Oxidizing substance — a material that intensifies combustion

Q43. A technician is completing a 1,000-hour service on an excavator. The hydraulic return filter has a restriction indicator showing RED (bypass mode). The filter element is changed. After restart, the indicator shows RED again within 5 minutes. What is the MOST likely cause?

- A) The replacement filter element is defective — install another new element
- B) Hydraulic system temperature is too low — cold oil is thick and causes false bypass indication
- C) Heavily contaminated fluid is rapidly loading the new element
- D) The bypass spring tension is too low — adjust the bypass relief

Q44. A machine's CAN bus (Controller Area Network) has a fault code indicating loss of communication with the transmission control module (TCM). The TCM powers up normally when tested independently. What is the MOST likely cause?

- A) TCM internal control circuitry has failed
- B) A wiring fault in the CAN H/CAN L pair at the TCM node
- C) ECM has failed — it controls all CAN communication
- D) Termination resistor at the TCM is defective

Q45. A technician uses oil analysis to monitor a diesel engine. The report shows TAN (Total Acid Number) of 4.0 mg KOH/g compared to a new oil baseline of 1.2 mg KOH/g. The oil is at 75% of change interval. What should the technician do?

- A) Continue to the full change interval — TAN of 4.0 is within normal range
- B) Add an oil neutralizer additive to reduce TAN
- C) Increase the change interval — high TAN indicates the oil is still absorbing acids effectively
- D) Change oil immediately — elevated TAN indicates acid buildup that causes corrosive wear

Q46. After rebuilding a hydraulic brake caliper, the brake drags during initial operation. The caliper pistons were replaced and the slides were lubricated. During diagnosis, the brake hose is found to be deteriorated internally. How does a deteriorated brake hose cause brake drag?

- A) A deteriorated hose restricts brake fluid flow, reducing hydraulic pressure to the caliper and causing partial engagement
- B) A deteriorated hose allows air ingestion into the system, creating a partial air lock that holds the caliper pistons applied
- C) Internal hose deterioration reduces hose flexibility, causing it to transmit vibration that continuously applies the brake pedal
- D) The collapsed inner lining acts as a one-way check valve — fluid enters under pedal pressure but cannot return

Q47. An operator has just connected a bucket to an excavator's hydraulic quick coupler. According to accepted safe-work guidance, what must be done before beginning work near ground workers?

- A) Verify lock engagement and load-test the bucket at ground level
- B) Raise the bucket overhead so the locking pins can settle
- C) Relieve auxiliary pilot pressure so the safety springs close
- D) Cycle the coupler switch several times to seat the wedge

Q48. While diagnosing a no-cooling complaint, a technician considers jumpering the low-pressure switch connector to force the compressor clutch to engage on a system reading well below normal charge. What is the main risk of running the compressor this way?

- A) Condenser overpressure from unrestricted refrigerant circulation
- B) Compressor seizure from oil starvation at low refrigerant flow
- C) Evaporator icing from continuous uncontrolled clutch engagement
- D) Clutch coil burnout from bypassing its resistor circuit

Q49. A powershift transmission shifts between gears roughly and with a noticeable engagement shock. The MOST likely cause is:

- A) Worn clutch packs or calibration out of specification
- B) High transmission operating temperature
- C) Low transmission oil level causing gear slip
- D) Incorrect oil viscosity — too thin oil causing hydraulic pressure loss

Q50. After installing a wheel on an articulated dump truck, the technician torques the nuts to specification in a star (criss-cross) sequence. The required follow-up is to:

- A) Re-check the nuts only at the next 250-hour service
- B) Re-torque the nuts after the first short period of operation
- C) Loosen and fully re-seat the wheel after the first shift
- D) Coat each stud with thread-locking compound while warm

Answer Key & Explanations

Q1. Answer: B

Slow crank = high resistance in the cranking circuit. The starter draws 200–800A. Even small resistance in cables or connections causes a significant voltage drop. Perform a voltage drop test on both positive and negative cables while cranking.

Q2. Answer: A

Intake → Compression → Power → Exhaust. During intake, air fills the cylinder. Compression heats the air to ~500°C+. Diesel injected at TDC auto-ignites (power stroke). Exhaust gases are expelled. Diesel engines have no spark plugs — compression alone ignites the fuel.

Q3. Answer: B

Terminals 85/86 are the coil — a reading in the 50–120 ohm range means the coil is intact, but says nothing about the load contacts. To finish the test, energize the coil with fused jumper leads (battery voltage across 85 and 86), listen for the click, then check continuity from terminal 30 to 87: near zero ohms energized, open circuit de-energized. Burned or pitted contacts fail this half of the test even with a perfect coil. Also inspect the socket terminals for corrosion or spread female contacts before condemning the relay.

Q4. Answer: A

PC pump standby: full compensator pressure at near-zero flow is normal. The compensator strokes the pump to minimum displacement while maintaining the set pressure (deadhead), ready for instant response when a function is activated. Energy loss at standby is small because flow is near zero, though standby pressure is high. A LOAD-SENSING pump, by contrast, drops to a low standby (margin) pressure — typically 200–400 PSI — when no function is active. Confusing the two leads to false "compensator failure" diagnoses.

Q5. Answer: A

Transfer case not fully engaged in low range = machine in high range = insufficient engine braking. Low range provides a lower (numerically higher) gear ratio that significantly increases engine braking effect. If the transfer case did not complete the shift, the machine remains in high range with much less engine braking. Verify the selected range using the service tool and check the range shift synchronizer. Also verify: retarder is fully functional, transmission is in the correct low gear.

Q6. Answer: B

Hammerless retention removes the hammer, which removes the struck-by hazard. Driving hardened pins with a sledge risks flying steel chips, missed strikes, and hand injuries. Hammerless systems use a threaded or lever-actuated retainer turned with a simple hand tool, so tooth changes are faster and safer. Wear life and inspection requirements are unchanged by the retention method.

Q7. Answer: C

Coolant-side liner pitting = cavitation erosion — prevented by maintaining nitrite-based SCA levels. Each combustion event makes the wet sleeve vibrate like a bell. As the liner wall snaps away from the coolant, vapour bubbles form; when it snaps back, the bubbles implode against the surface and hammer out pits that can eventually perforate into the cylinder. Nitrite in the supplemental coolant additive (SCA) package forms a sacrificial oxide film that absorbs the implosions. Test coolant with SCA strips at PMs and maintain roughly 1.5–3.0 units per gallon, or use an extended-life coolant per OEM.

Q8. Answer: D

ISO 4406: three numbers = particle count ranges at $\geq 4\mu\text{m}$, $\geq 6\mu\text{m}$, $\geq 14\mu\text{m}$. Higher number = dirtier oil. Each increment of 1 in the ISO code represents doubling the particle count. Code 22 means 20,000–40,000 particles per mL at $\geq 4\mu\text{m}$. Servo valves have bore clearances of just 1–5 μm — even particles much smaller than visible can cause valve erosion and stiction, so servo systems typically require ISO 16/14/11 or cleaner. Typical cleanliness requirements: servo valves = ISO 16/14/11; piston pumps = ISO 17/15/12; gear pumps = ISO 20/18/15. Code 22/20/17 would rapidly destroy precision components.

Q9. Answer: D

Compressor builds to 120 psi correctly, but system drops to 80 psi triggering the warning — this is a downstream leak, with air escaping faster than the compressor maintains when stationary. The compressor output is confirmed good (builds pressure, cuts out at spec). The pressure drop after cut-out is caused by air escaping through a leak in the system (brake chambers, supply lines, fittings, drain valves). Perform a static leak test: engine off, brakes released, monitor pressure drop rate. Should not exceed 2 psi/minute for service brakes.

Q10. Answer: B

HEUI: high-pressure oil (1,000–3,500 PSI) actuates an intensifier piston to generate injection pressure. The HEUI injector uses engine oil at 1,000–3,500 PSI to push an intensifier piston with a 7:1 ratio, generating fuel injection pressure of 3,000–21,000 PSI. Without actuation oil pressure, the intensifier cannot move and the injector cannot generate the high fuel pressure needed for injection — fuel pressure stays at low transfer pump pressure (~60–80 PSI), completely insufficient. Diagnosis: check high-pressure oil pump, IAPCV solenoid, and oil circuit integrity. Common on Cat 3126/C7 and early International DT engines.

Q11. Answer: B

Air in a hydraulic brake circuit is compressible — unlike brake fluid — causing spongy, high-travel pedal. Air can enter through: micro-leaks in connections (suction side), air bubbles from aeration (low fluid level at some point), or improper bleeding procedure. Systematic bleeding from the furthest wheel cylinder to the nearest purges air. If bleeding does not correct, suspect a faulty master cylinder seal allowing air to be drawn back in.

Q12. Answer: D

Low charge pressure = hydrostatic drive weakness in both directions. The charge pump maintains circuit pre-charge pressure (typically 200–400 PSI) to prevent cavitation and replenish leakage. If charge pressure is low, the main pump cannot develop full displacement in either direction. Symptoms: weak in both forward and reverse, no fault codes on simpler systems. Check charge pressure with a gauge on the charge port.

Q13. Answer: B

Burning magnesium (Class D): ONLY a Class D agent rated for metal fires — never water, CO₂, or ABC dry chemical. Water reacts violently with burning magnesium, producing hydrogen gas that can explode and splatter molten metal. CO₂ is ineffective and may react. ABC dry chemical does not suppress metal fires. Class D agents (graphite powder, copper-based powder, dry sand) smother by forming an insulating crust without reacting with the metal.

Q14. Answer: B

Dither = small high-frequency AC signal superimposed on the DC control signal, preventing valve spool stiction. Precision valve spools have very tight tolerances (clearance measured in microns). Static friction (stiction) can cause the spool to stick between signal changes, resulting in jerky, non-linear, poorly controllable response. Dither — typically 50–400 Hz, small amplitude — keeps the spool in constant microscopic motion, eliminating stiction. Without it, the valve has a "dead band" where it doesn't respond until force overcomes static friction, then jumps. Effect on machine: without dither, boom/arm movement becomes jerky especially at fine control inputs.

Q15. Answer: C

Wet disc brake drag: residual apply pressure or failed return spring preventing full piston retraction. Wet disc brakes use separate apply and cooling oil circuits. After the brake pedal is released, the apply circuit must fully depressurize to allow the piston to retract. A trapped pressure pocket (check valve, restricted return line, or pressure metering valve not returning to zero) keeps the piston partially applied and the discs partially clamped. A damaged or missing return spring cannot push the piston back. Test: measure apply circuit pressure at brake with pedal fully released — should be 0 PSI. Also measure brake cooling oil temperature — drag generates excess heat.

Q16. Answer: A

Sweet smell plus oily window film plus a wet floor points to coolant from the heater core. Glycol vapour from a leaking heater core condenses as a greasy film on the glass, and liquid coolant drips onto the floor below the HVAC housing. A leaking evaporator loses refrigerant and oil, not coolant, and produces poor cooling rather than these symptoms.

Q17. Answer: A

EGR: introduces cooled exhaust gas into the intake, lowering combustion temp = lower NO_x. NO_x (nitrogen oxides) forms at extremely high combustion temperatures. Recirculating inert exhaust gas into the intake dilutes the oxygen content and absorbs heat, reducing peak combustion temperature and NO_x formation. EGR is part of the EPA Tier 4/Euro 6 emissions compliance strategy along with SCR/DEF.

Q18. Answer: C

Attachment matching starts with flow (L/min) and pressure (kPa/psi) against the carrier's auxiliary circuit specs. Too little flow makes the tool slow or stall; too much flow overspeeds motors and overheats the tool, while a mismatched relief setting either starves the tool of force or overpressurizes it. Manufacturers publish required flow and operating pressure for every attachment, and the carrier's auxiliary output must fall within that window. Secondary items such as case drains apply only to specific tool types.

Q19. Answer: A

Accumulator = pressurized fluid storage that supplements pump flow on demand. A bladder or piston-type accumulator uses nitrogen pre-charge to store hydraulic energy. On peak demand, it releases stored fluid to assist the pump. Also used for dampening pressure spikes and maintaining pressure during brief pump shutdowns.

Q20. Answer: C

Before welding on a machine: disconnect the battery cables, and clamp the work lead as close to the weld as possible. Welding current returns through the path of least resistance — a work clamp placed far away can send hundreds of amps through pivot pins, bearings, and module ground circuits, destroying ECMs and arcing bearing surfaces. Many OEMs additionally require unplugging ECM connectors or removing modules mounted near the weld area. Never run the engine while welding.

Q21. Answer: A

Final drive: reduces transmission output speed and proportionally multiplies torque at the wheels or sprockets for maximum pulling force. Heavy equipment requires enormous ground-level torque that transmissions and torque converters alone cannot produce efficiently. The final drive (planetary gear set inside each wheel hub or sprocket drive) provides the last stage of speed reduction and torque multiplication. Ratio typically 10:1 to 30:1 depending on equipment.

Q22. Answer: B

Intermittent faults are almost always wiring/connector issues causing an intermittent open circuit, not component failures. A failed sensor typically produces a continuous fault — it works or it doesn't. An intermittent signal that changes with vibration, temperature, or movement indicates a chafed wire, corroded connector pin, or broken wire strand that makes intermittent contact. Check the harness for damage near moving components and clean/reseat the connector.

Q23. Answer: C

Force = Pressure × Area. $\text{Area} = \pi \times r^2 = 3.14159 \times (2)^2 = 12.57 \text{ in}^2$. $\text{Force} = 2,000 \times 12.57 = \mathbf{25,133 \text{ lbs}}$. This relationship explains why hydraulic systems generate enormous forces from compact components.

Q24. Answer: A

Normal pressure + slow speed = flow restriction. If the pump were worn, pressure would be low too. A blocked hydraulic filter or restricted flow control valve limits oil volume (GPM) reaching the cylinder, slowing movement without affecting system pressure reading.

Q25. Answer: A

60 ohms = two 120-ohm terminating resistors in parallel — the expected reading for a properly terminated CAN bus. Each end of the backbone carries a 120-ohm resistor; measured together in parallel they give $(120 \times 120) / (120 + 120) = 7200 / 240 = 60 \text{ ohms}$. A reading of about 120 ohms means one terminator is open or missing; well below 60 ohms indicates a short between CAN High and CAN Low. Always measure with the network powered off.

Q26. Answer: B

Accumulator = pressurized fluid reservoir. Uses a gas charge (nitrogen) separated from oil by a bladder or piston. Functions: 1) Stores energy to supplement pump flow during peak demand (faster cylinder response), 2) Maintains brake/steering pressure after engine shutdown, 3) Dampens pressure spikes. NEVER cut open an accumulator — high-pressure nitrogen is dangerous.

Q27. Answer: B

Turbo lag: black smoke at light load, clears under hard acceleration. At low loads and RPM, the turbo hasn't spun up to produce adequate boost. The engine receives excess fuel relative to available air — resulting in a rich/incomplete combustion (black smoke). As load increases, exhaust energy spins up the turbocharger, boost pressure rises, and the air-fuel ratio normalizes — the smoke clears. This is normal behaviour but excessive lag can indicate a worn or damaged turbocharger.

Q28. Answer: B

Out-of-calibration torque wrench: remove from service — it can leave critical fasteners loose or over-tightened. Torque wrenches should be calibrated at least annually or after being dropped. An offset of $\pm 4\%$ is typically the maximum acceptable tolerance. At 10% error, critical fasteners (cylinder head bolts, wheel hub nuts, connecting rod bolts) may be under-tightened and fail. Send for factory recalibration or replace.

Q29. Answer: B

Differential allows speed difference between drive wheels when turning. On a turn, the outer wheel travels farther than the inner wheel and must rotate faster. Without a differential, tires would scrub and drivetrain stress would be extreme. The differential uses bevel gears to split torque while allowing different wheel speeds.

Q30. Answer: B

Wire ampacity: lower gauge number = thicker wire = higher ampacity. AWG 14-gauge is rated for approximately 15 amps maximum in chassis wiring. A 25-amp circuit exceeds this by 67%. Undersized wire develops resistance heating proportional to I^2R — at 25A through 14-gauge, the wire will overheat, melt and damage insulation, and create a fire and circuit failure hazard. The original 10-gauge wire is rated for approximately 30 amps continuous. Always match or exceed the original wire gauge in repairs. Approved repair: use 10-gauge or 8-gauge (heavier) wire.

Q31. Answer: C

Excessive back pressure + black smoke + power loss under load = plugged DPF (Diesel Particulate Filter). A clogged DPF increases exhaust restriction, reducing the engine's ability to expel burned gases. This causes power loss, elevated exhaust temperatures, and increased emissions. Solution: perform a forced DPF regeneration or remove and clean/replace the DPF.

Q32. Answer: B

Aeration vs Cavitation: Aeration = air mixed into oil (foamy, milky appearance, spongy operation — from leak on suction side or low fluid). Cavitation = vapor bubbles formed by low suction pressure (high-pitched whine, pitting/erosion damage on pump internals).

Q33. Answer: D

Dynamic retarding dumps braking energy as heat in resistor grids. On grade, the wheel motors switch to generator mode and the electrical power they produce is routed to large resistance grid elements, where a blower carries the heat away. This slows the truck with minimal service-brake application, extending brake life. Machines with hybrid storage can capture some of this energy instead, but a conventional electric-drive truck dissipates it.

Q34. Answer: B

Ring input + carrier held + sun output = speed increase + direction reversal. The planets walk around the stationary carrier and drive the sun gear in the opposite direction at a multiplied speed. Ratio = $1 + (R/S)$, but since sun is output: sun spins faster than ring AND in reverse. This is used in reverse gear on some transmissions.

Q35. Answer: C

In Canada, only a certified person may recover or handle halocarbon refrigerants. Federal and provincial halocarbon regulations require technicians to complete an approved environmental awareness / refrigerant handling course before servicing AC systems or recovering refrigerant. The training covers leak detection, approved containers, and recovery, recycling and reclamation practices.

Q36. Answer: D

Low boost has three main causes: 1) Restricted air intake before the turbo (dirty filter), 2) Boost leak in the charge air system after the turbo (cracked pipe, loose clamp), 3) Worn turbocharger (blade wear, shaft play). Check in this order.

Q37. Answer: C

Clutch slip = RPM up, speed not up proportionally. A slipping clutch pack fails to transfer engine torque to the output shaft. The engine revs freely while the machine barely moves — similar to a slipping manual clutch. Burnt oil smell and overheating transmission oil are also signs.

Q38. Answer: C

Normal current but slow crank = voltage not reaching the motor. High cable resistance causes an excessive voltage drop to the starter. The starter draws its rated current, but at reduced voltage (8V instead of 12V), it produces much less power and torque. Perform a voltage drop test on both positive and negative cables while cranking.

Q39. Answer: A

Black smoke = rich mixture (excess fuel, insufficient air). White smoke during warm-up is normal (unburnt fuel/water condensation) and clears when hot. Black smoke at full load with serviceable fuel system points to air starvation — the air filter is restricting intake. The engine can only burn as much fuel as available air allows. Check air filter restriction indicator; replace if above spec.

Q40. Answer: D

Quick-release valve exhausts air locally at the brake chamber. When the driver releases the brake pedal, air must exhaust. Without a quick-release valve, air would have to travel all the way back to the foot valve to exhaust — slow release means slow brake release (brake drag). The quick-release valve vents air at the chamber immediately.

Q41. Answer: D

Open-center: continuous flow through the valve centers back to tank at low pressure when valves are in neutral (pump always spins fluid). Closed-center: pump output stops (variable displacement) or pressure is held at standby when no function is active — more energy efficient at idle. Open-center control valves allow oil to flow from pump → through the valve center → back to tank when in neutral. The pump always delivers flow (fixed displacement), consuming engine power. Closed-center valves block flow in neutral — with a variable displacement pump, the pump destrokes to minimum displacement (low energy use). This is why modern high-end equipment uses closed-center with load-sensing systems.

Q42. Answer: D

Flame over circle = oxidizer (GHS). The plain flame symbol indicates flammable materials. The flame over a circle specifically indicates oxidizing gases, liquids, or solids — materials that can intensify combustion and cause or contribute to the burning of other materials. Oxidizers supply oxygen to fuel a fire — they make fires burn hotter and are harder to extinguish. Examples: hydrogen peroxide, liquid oxygen, ammonium nitrate. Must be stored away from flammable materials.

Q43. Answer: C

A new filter element bypassing immediately after installation indicates the hydraulic fluid is heavily contaminated — the new element immediately loads up with system contamination. The filter captures contaminants from the fluid. If the fluid is severely contaminated (from a pump failure, maintenance contamination, or long overdue change), the new element fills rapidly. Solution: flush the system (tank drain, all cylinders and motors cycled, filter changed multiple times), replace the fluid, and identify the contamination source.

Q44. Answer: B

TCM works independently but not on CAN = communication wiring fault (open or short in the twisted pair between the TCM node and the bus backbone), not TCM failure. CAN bus uses twisted pair (CAN H and CAN L). Faults: broken wire (open), shorted wires, damaged shield, missing termination resistor. Test: measure CAN bus resistance with ignition off (should be ~60 Ω for a properly terminated bus). Incorrect resistance = missing/failed termination resistor. Use a CAN bus analyzer or oscilloscope to find break location.

Q45. Answer: D

High TAN indicates acid accumulation in the oil from combustion byproducts and oil oxidation. New oil TAN: ~1.2 mg KOH/g. A TAN of 4.0 mg KOH/g at 75% interval indicates the oil has lost its alkalinity reserve (TBN) and is now acidic. Acidic oil causes corrosive wear on engine bearings and cylinder walls. Change oil immediately and investigate root cause (overextended intervals, blowby contamination, fuel dilution).

Q46. Answer: D

Deteriorated brake hose: collapsed inner lining acts as a check valve — fluid goes in but can't return, so the pistons remain applied after the pedal is released. The outer appearance of a brake hose can look fine while the inner lining delaminates and collapses. Under pedal application, pressure pushes past the blockage and applies the brakes. When the pedal is released, the return path is blocked by the collapsed lining — the caliper remains applied. Test: with brakes applied, crack the caliper bleeder valve — if the piston releases immediately when bleeder is opened, the restriction is in the hose.

Q47. Answer: A

Every coupler connection must be verified visually and by a ground-level load test before work starts. Most dropped-bucket incidents happen because the attachment was never fully engaged and locked, even though it appeared connected. Confirming the lock indicator or pin position and crowding the bucket against the ground proves engagement under load away from workers. Manufacturer connection-test procedures must be followed every time an attachment is changed.

Q48. Answer: B

Refrigerant carries the oil, so running a low-charge system starves the compressor of lubrication. The low-pressure switch exists precisely to prevent this: with little refrigerant circulating, oil stops moving through the system and the compressor can score and seize. Jumpering the switch is acceptable only very briefly as an electrical diagnostic, never as a way to operate a low system.

Q49. Answer: A

Rough powershift: clutch calibration (fill time, engagement pressure) or pack wear causing abrupt engagement instead of smooth overlap. Powershift transmissions shift by precisely overlapping clutch packs — one releases as the other engages. If calibration data is off (fill time too short, engagement pressure wrong) or clutch packs are worn thin, the overlap is poor and shifts are abrupt. Electronic transmissions can often be recalibrated via service software.

Q50. Answer: B

New wheel installations must be re-torqued after a short break-in period. As the mating surfaces of the hub, wheel, and nuts seat under load and thermal cycling, clamping force can relax below spec, which is how wheel-off incidents start. Manufacturers typically call for a re-torque within roughly the first 10 service hours or 50 to 100 km of operation, then regular checks afterward. The star sequence ensures the wheel seats evenly and concentrically at every torque pass.

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